

AMME5710 Major Project: Live Tennis Player & Ball Tracking Pipeline

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Abstract

Live tennis Video Assistant Referees (VARs) and analytics have traditionally relied on multi-camera systems such as Hawk-Eye, which employs 10–12 high-speed cameras to triangulate ball trajectories and player positions with millimetre precision. While effective, such systems require several days of setup and calibration and cost upwards of USD 100,000 per court in hardware, rendering them inaccessible for lower-tier competitions. This project addresses the need for a low-cost, easily deployable alternative capable of extracting meaningful match statistics directly from broadcast footage.

We design an end-to-end computer vision pipeline that operates in near real time on standard 25 FPS television footage to detect and track both players and the ball, compute positional and motion-based metrics, and output a visualisation of the game from a top-down perspective. The system is intentionally lightweight, relying primarily on classical computer vision methods for court detection, homography estimation, foreground segmentation, and object tracking, with learning-based components used only where necessary to validate the classical models.

Tested across all four Grand Slam tournaments, spanning differing camera angles, court colours and surface textures, our method achieves robust and consistent player tracking and reliable court calibration. Ball tracking performance is more limited, although rally trajectories can often be recovered, the use of a single monocular camera prevents recovery of ball height, and the low frame rate and motion blur of fast serves and ground-strokes cause frequent trajectory fragmentation and unreliable or missed detections. Nonetheless, the results demonstrate that player-position analytics and approximate ball trajectories can be derived from broadcast video without specialised hardware, suggesting a viable pathway towards affordable tracking and analytics tools for lower-tier and amateur tennis.

Our code base can be found on Github <https://github.com/sduckworth234/ComputerVisionMajorProject>

1. Introduction

State-of-the-art tennis analytics systems like Hawk-Eye use multi-camera set ups with high-speed sensors and precise calibration to triangulate ball and player trajectories with millimetre accuracy [5]. While effective, these setups are costly and complex, making them inaccessible to lower-tier tournaments, training facilities, and amateur players.

This project explores whether comparable tracking data and match statistics can be extracted from a single monocular broadcast feed. Television footage poses challenges such as camera pans, zooms, cuts, variable lighting and colour shifts. However, if proved successful, existing footage eliminates the need for additional invasive sensors or player wearables. By using classical computer vision techniques our modular pipeline aims to automatically recover player positions, ball trajectories, and court geometry, with learning-based methods used only as benchmarks.

2. Literature Review

2.1. Court Detection & Calibration

There has been recent literature that presents more enhanced tennis court line detection algorithms which are capable of handling irregular court markings. These models [2] extend traditional Hough-line detection with deep learning components to assist. These are pre-trained shadow removal models and net detection to find coordinate boundaries to crop the image, this specific approach aims to be more robust and faster for lower quality footage which comes from amateur game recordings. From these simple “helper” modules, this model achieves a more stable homography estimation and reliable line detection for varied light and surface conditions. However, this simplification fails to capture 3D geometric accuracy which is explored in another method [7] across multi-view broadcast feeds. The data for this pipeline is tracking soccer pitch layouts and player positions, camera parameters such as focal length, it is to be noted that extrinsics vary significantly in this input data. To combat this, the model utilised the structural layouts of sports fields and redefined the problem space as a geometric problem rather than a homography fit. This [7]

allows for accurate tracking of positions using HRNetv2 for heatmap feature localisation and Direct Linear Transformation (DLT) with RANSAC. As a result, this approach achieves both single and multi view camera calibration.

2.2. Player Tracking

Player tracking in tennis is typically framed as an object detection and multi-object tracking problem. Existing approaches range from classical background subtraction and image segmentation through to deep learning-based detectors such as YOLO and Regional Convolutional Neural Networks (R-CNN). Pose estimation frameworks and action recognition [3] have also been used for more accurate tracking. The integration of pose-based methods with trajectory analysis and temporal modelling has enabled more robust tracking under rapid motion and partial occlusion, and can be used to predict future player movement using joint locations, player trajectories and ball position [3].

Accurate player tracking generally begins with reliable player detection via bounding boxes or similar region proposals. YOLO and its variants are well suited to real-time sports analytics because they offer fast, single-shot detection of multiple players and objects in each frame [4]. Two-stage detectors such as R-CNN usually provide higher detection accuracy and more precise segmentation, but at the cost of substantially slower inference, making them more appropriate for offline or highly specialised tasks.

2.3. Ball Tracking

Ball tracking has various approaches similar to player tracking using a combination of traditional and ML computer vision processing methods. Image processing based techniques such as frame differencing, background subtraction, optical flow and Kalman filtering form the backbone of many traditional ball tracking pipelines [9]. An advantage of this approach is that it is highly modular, making it adaptable and allows for scalability to improve accuracy or robustness to differing environments. An example of this is utilising bounce and spin physics to more accurately track the ball without computational overhaul [11]. Physics based models offer robustness and uplift for real-world conditions such as occlusion and variable lighting as predictions are not solely reliant on vision processing.

A salient, commonly used alternative is to employ CNN's for ball detection combined with YOLO based models [8]. These models are prominent for their speed and robustness in detecting small, fast moving objects in cluttered environments. Supervised learning is leveraged to recognise balls directly from image frames, overcoming thresholding limitations for traditional methods (ability to withstand background noise and motion blur). DeepSORT and other multi-object trackers are used to link successive ball detections to produce continuous trajectories, a study [10]

fine tuned YOLO for enhanced tennis ball detection tailored for the unique rapid movement that is inherent to tennis.

2.4. Project Constraints & Opposing Research Gap

With the following assignment constraints and deep focus on traditional computer vision methods as much as possible, we want to build a vision pipeline that can be easily improved by a mix of ML algorithms within. Our pipeline focuses on modular integration and interpretability rather than algorithmic novelty. A lean traditional approach that we are looking to implement opposes the research gap of having a specific pipeline, our pipeline allows for a "minimum viable product" where ML models can be placed and tested where needed. We propose that this is the least worst option given key stakeholders i.e. the project brief, unit outline and marking criteria.

NB: In one module we implement MOG2 (ML based).

3. Methodology

3.1. Overview

We built a modular computer vision pipeline that processes raw, single-camera broadcast tennis footage. Through successive stages of court calibration, player detection/tracking and ball candidate association we aim to output a geometrically consistent top-down representation of the match. From this calibrated view we compute meaningful baseline statistics such as player and ball trajectories through lightweight, classical image-processing methods and using small learned components to complete the pipeline.

3.2. Key Requirements

The system must use only a single broadcast camera with the pipeline remaining stable despite typical broadcast artefacts. As well as this, the detection and tracking of corners must remain stable in camera pans and zooms.

3.3. Assumptions

Our court classifier assumes a standard broadcast camera positioned behind the courts baseline with visible court markings and no camera motion. Court lines must remain visible for reliable corner localisation, and the frame sufficiently static for MOG2-based segmentation. We also assume a singles game with two players at either end of the court at all times and a single ball.

3.4. System Architecture & Design Approach

Tennis presents a complex visual environment characterised by dynamic lighting conditions, fast player and ball motion, and frequent interruptions such as replays or crowd shots. Broadcast footage is particularly challenging, as it

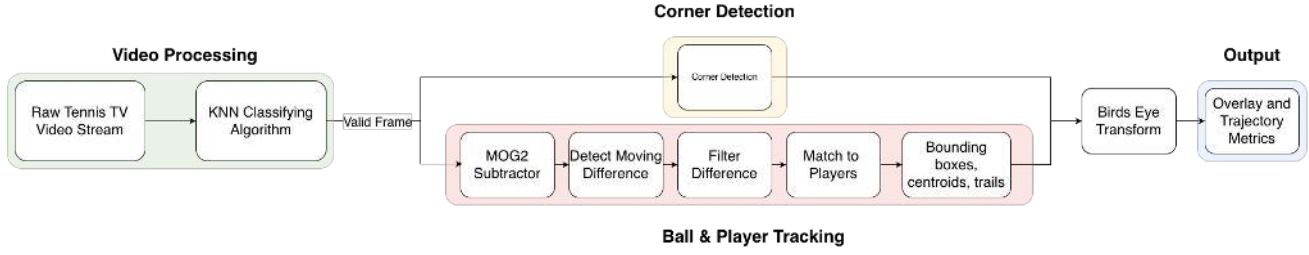


Figure 1: Image Processing Pipeline

introduces motion blur, camera pans, and scene transitions that complicate consistent analysis.

To address this, the project is composed of five sequential components, video processing, court and corner detection, ball and player tracking, transformation, and output visualisation.

As seen in Figure 1, the system begins with video pre-processing, where incoming frames are classified as either game play or auxiliary footage using a classifier. Valid frames then follow two parallel paths, court and corner detection, responsible for establishing a stable geometric reference frame by identifying key court landmarks, and secondly, ball and player tracking, which isolates and follows moving objects of interest. The results of both modules are merged in the transformation stage, where detections are projected into a bird’s-eye view using a computed homography transformation. The final stage renders these results, providing visual tracking overlays.

3.5. Tennis Frame Classification

To operate in real time on live broadcast footage, the first stage of the pipeline filters the stream to retain only game play frames from the main baseline camera. Frames from other angles or where spectators or overlays block a significant portion of the court, are classified as invalid and discarded.

We implement a lightweight custom KNN-based frame classifier using a compact, interpretable 10-dimensional feature vector produced by our feature extraction methods, designed to encode whether a frame is a full-court view. The most salient structure is the net, captured via a *horiz_band_energy* feature, the frame is converted to grayscale, vertical gradients are computed using a Sobel operator, and the total *edge energy* in the central 40% horizontal band (where the net typically lies) is compared to the rest of the image.

To detect court markings, the *white_frac* feature estimates the proportion of white line pixels. The frame is converted to HLS colour space, thresholded on lightness and saturation to produce a colour-invariant white-line mask, and then morphologically closed to fill small gaps; the fraction of white pixels provides a strong cue for full-court views, which consistently exhibit prominent lines. The linear structure of the scene is further quantified using

OpenCV’s Line Segment Detector. For each detected segment we compute its length, discard short spurious lines, and calculate statistics such as the number of long segments, mean length and length standard deviation, reflecting the presence of long, straight, parallel court lines.

Edge orientations provide an additional measure of geometric order. From the detected segments we compute orientation angles and summarise their distribution to distinguish structured court geometry (dominant horizontal and vertical directions) from more random non-game play scenes. Finally, an image sharpness metric acts as an auxiliary feature, under the assumption that game play frames are relatively sharp and stable, whereas replays, transitions and camera motions often appear blurred. Together, these features allow the KNN classifier to robustly separate valid baseline game play frames from other broadcast content, ensuring that subsequent modules operate on footage consistent with our problem assumptions.

3.6. Court Corner Detection & Localisation

3.6.1 Classical Approaches

Initial approaches involved canny edge detection followed by a hough line transform to extract straight line candidates corresponding to court markings. To identify centroid clusters near the image centre (court lines) connected components was used. Subsequent cropping around the largest centroid near the centre was used to remove any other masked figures. A simple maxima minima post-processing approach was implemented to consistently label detected court corners as top-left, top-right, bottom-left, and bottom-right. Each point’s spatial position was analysed using its coordinate sums and differences. Specifically, for each detected point (x, y) , the values $x + y$ and $x - y$ were calculated. Points with the smallest sum $(x+y)$ correspond to the top-left of the image, while those with the largest sum represent the bottom-right yet this proved unsuccessful as discussed in the results section.

Seen in Figure 2, the final classical attempt employed OpenCV’s Line Segment Detector (LSD) to identify line segments, which were grouped by orientation into near-horizontal baselines and near-vertical sidelines that were within frame regions. These lines were extended and the intersections were used to approximate court corners.

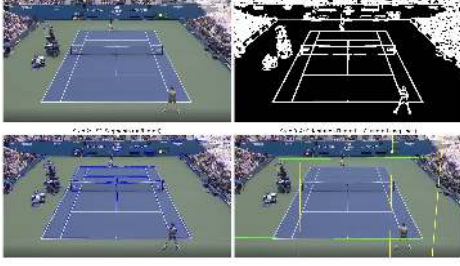


Figure 2: Line Segment Detector

3.6.2 Hybrid Model

To overcome these roadblocks and subsequent pipeline limitations, we used a "plug and play black box" lightweight court key point detector trained to identify 14 standard tennis-court landmarks. We modified the model to extract only the 4 perimeter corners. This model is a pre-trained custom convolutional neural network that was trained to predict encoded heat maps of tennis-court landmarks. From this, each key-points peak gives a (x_pred, y_pred) position in the full resolution frame. To refine accuracy, this model utilises classical vision methods where there is a local crop of a 40x40 window around the predicted point, isolating the nearby segments while ignoring the rest of the frame. Within each crop, `HoughLinesP()` is ran with every position drawn in red as seen in Figure 3. Next is merging, by sorting these segments replacing them with a single averaged segment. Finally the crossing point is solved for more accurate coordinates.



(a) Raw Lines (b) Merged Lines (c) Refactored Corner

Figure 3: Comparison of raw, merged and refactored corner line detections.

It is to be noted that if corner detection fails on a given frame, the pipeline reuses the most recent valid transform to preserve temporal continuity. While this may introduce gradual drift during long invalid sequences, it significantly reduces flicker and instability in the final visualisation and is a sacrifice for model robustness.

NB: This model lies beyond the course scope and was used only as a fallback "black box" to complete the pipeline.

3.7. Birds Eye Perspective Transformation

Using the detected court corners, a perspective transform maps the camera view to a canonical top-down court view.

The doubles court has real-world dimensions of length $L_{court} = 23.77$ m and width $W_{court} = 10.97$ m [1], giving a scale factor of $\alpha = \frac{L_{court}}{W_{court}} = \frac{23.77}{10.97} \approx 2.167$.

From the four detected corner points, the width of each baseline was calculated using the euclidean distance in the pixel domain. Due to the oblique angle of the camera, the court's top baseline appears shorter in pixels than the bottom baseline. As such, to ensure a balanced distortion correction, we used the average of these two measurements as the reference width for the warped space.

The warped image dimensions are then calculated to maintain the court aspect ratio, with 15% expansionary padding applied to ensure players were tracked even when moving outside the court perimeter by $h_{court} = w_{avg} \cdot \alpha$.

The homography matrix is computed using OpenCV's `getPerspectiveTransform` function and is applied to the raw frames, enabling tracked objects to be overlaid onto a consistent reference plane. The transform matrix is computed from source points (camera view court corners) to destination points (bird's-eye rectangle) and is applied on the first valid detection per rally to reduce jitter.

For clarity, to convert warped pixel coordinates to real-world metres, scaling factors are computed as:

$$s_x = \frac{W_{court}}{w_{avg}} = \frac{10.97}{w_{avg}}, \quad s_y = \frac{L_{court}}{h_{court}} = \frac{23.77}{w_{avg} \cdot \alpha} \approx s_x \quad (1)$$

(see Appendix 2)

3.8. Player Tracking

For player tracking in each frame, players are segmented using OpenCV's Mixture of Gaussians (MOG2) background subtraction model, using a history window of 300 frames. By trial and error this allows the system to isolate foreground objects and ultimately mask out unwanted artefacts, regions and noise.

Morphological operations were applied to further refine the foreground mask, specifically, a 3x3 opening kernel is applied to remove small fragmented regions which is then followed by a dilation and 7x7 closing kernel to fill small gaps and reconnect players limbs together forming a complete region of interest. To further filter and eliminate non-player objects, we filtered based on area between 750-30,000 pixels to accommodate the two players being at varying distances from the focal point. Additionally aspect ratio masking was applied under the assumption that a player bounding box will be taller than it is wide.

The final result outputs 2 bounding boxes around each player, taking the largest y value of each and middle x coordinate such to track their lowest centre of mass position. This was done to ensure that when mapping to the birds eye domain, each players position resembled where they were standing on the court. To increase accuracy, each player is

tracked over time using a simple Kalman filter to predict future positions based on prior motion states. Frame to frame association uses a combination of absolute centroid distance difference thresholds and Intersection Over Union (IOU) scores to maintain identity across frames. Player identification is initialised by splitting the frame horizontally, with the upper and lower halves assigned to opposing players.

3.9. Ball Tracking

For ball tracking, an initial attempt used HSV thresholding. However, this was unreliable under changing lighting as the tennis ball, while inherently the same colour it was visually different across different datasets. To improve robustness, we implemented a motion-based masking approach leveraging the player tracking methods of frame differencing and background subtraction with stricter geometric and temporal constraints due to the balls small size, high speed, occlusion and susceptibility to motion blur.

Firstly, each frame is contrast enhanced to increase the visibility of small fast moving objects against the static court background. We compare the previous two frames to generate a motion map to highlight fast-moving regions while ignoring slow background changes. Morphological opening and closing was used to remove small artefacts and retain compact, ball-like regions.

To prevent confusion with players, player masks are subtracted from the motion map, removing any overlapping movement caused by limbs or racquets. This is to improve robustness in our ball tracking model to sacrifice times where the ball is not tracked.

Similar to players, a Kalman filter (position and velocity) was used to smooth ball trajectories and predict positions during missed detections, with the parameters tuned to handle the ball's unpredictable motion, updating its state with each detection. Additionally, the extracted contours are tested for size (10-120 pixels squared) and shape (circularity) and position (outside player boxes). Each valid candidate is scored using geometric and physical consistency metrics as well as its distance from its Kalman filter prediction. Finally the candidate with the highest overall score is chosen as the valid detection for that frame.

4. Experimental Design

4.1. Frame Classification Accuracy

To benchmark the frame classifier, we constructed a labelled dataset from men's and women's Grand Slam finals highlight videos (Australian Open, French Open, Wimbledon and US Open) downloaded from YouTube. Using a small extraction script, every second frame was saved as a PNG image to reduce redundancy while preserving temporal variation, yielding 92,970 frames (~one hour of footage). Each frame was manually annotated as either a

valid gameplay frame (main camera, full court visible) or an invalid frame (close-ups, alternate angles, heavy occlusions). This ensures that the evaluation set covers different court surfaces, colours and lighting conditions and is not biased towards a single tournament or gender.

Class balance was enforced by downsampling the majority class so that valid and invalid examples contribute equally to training and testing. A KNN classifier was then trained on the extracted feature vectors, using the manual labels as ground truth. Logistic regression was explored as a baseline and achieved comparable performance, but KNN was ultimately preferred for its simplicity and low computational cost at inference time. Frame classification accuracy is measured as the proportion of correctly classified frames on a held-out test split of this dataset.



Figure 4: Sample valid frame



Figure 5: Sample invalid

4.2. Court Corner Detection Accuracy

For the classical approach initial testing included test frames of each approach compared labelled ground truth coordinates. Mean and standard deviation distance errors for frames for each corner were compared to the ground truth. Each of the environments to test were the Australian Open, US Open and Wimbledon to provide a wide range of court environments to determine feasibility.

For the plug-and-play hybrid model, a similar evaluation process was applied. Each test image was loaded alongside its corresponding key and metadata, with a file containing the ground-truth corner coordinates for that specific frame. The model's predicted corner locations were then compared directly against these ground-truth values. From these comparisons, summary statistics such as mean absolute error, standard deviation, and per-corner deviation were computed to quantify the model's accuracy and consistency across the dataset.

4.3. Player Tracking

We curated two datasets in our testing, a Wimbledon womens rally and a mens rally from the Australian Open, both of length 8 and 14 seconds respectively. Tracking performance was assessed via RMSE between predicted and YOLO positions, converted from pixel to metric coordinates using dataset-specific conversion factors.

4.4. Ball Tracking

To evaluate ball tracking accuracy, a labelled tennis ball dataset and manual ground truth annotation was used with a YOLO model we trained on a roboflow dataset [6]. The ball tracking process was run on the full video to identify high confidence frames where the top candidates exhibited high overlap and similar geometric properties. A subset of frames meeting the criteria were extracted and manually labelled as ground truth. This allowed a direct comparison between predicted and actual ball positions through pixel-wise error metrics.

5. Results & Discussion

5.1. Frame Classification Accuracy

Results show that the frame classifier reliably filters non-gameplay scenes, improving the stability of downstream modules. In an initial pilot on a small, balanced test set drawn from all eight Grand Slam final highlight videos ($N = 40$), the classifier achieved an accuracy of 97.5%, with precision of 96.0% and recall of 100%.

To obtain a more reliable estimate of performance, we then trained and evaluated on a much larger, balanced subset of the dataset. From 36,360 labelled frames (23,112 VALID, 13,248 INVALID) sampled across all eight finals, a 20% hold-out split yielded $N = 7,272$ test frames. On this larger test set the classifier achieved an accuracy of 99.5%, with precision of 99.4%, recall of 99.8% and F1-score of 99.6% (confusion matrix in Table 1). Only 37 of 7,272 frames ($\approx 0.5\%$) are misclassified, indicating that the feature set and KNN classifier generalise well across different tournaments, surfaces and broadcast conditions.

Table 1: Confusion matrix for frame validity classification on the large balanced hold-out test set ($N = 7,272$).

Actual	Predicted	
	VALID	INVALID
VALID	4615	9
INVALID	28	2620

However, the KNN classifier is reliable only for baseline views where the camera looks approximately straight down the court. As illustrated in Figure 6, even modest pans away from this baseline view can lead to misclassification, reflecting that most training frames come from a largely stationary viewpoint. This failure mode is most likely driven by our net-proxy feature (*horiz_band_energy*), which assumes that the net lies within the central 40% of the image height. When the camera pans or tilts such that the net moves out of this band, the frame appears more similar (in feature space) to an INVALID view. A more robust approach would dynamically update the net band by tracking the net position

over time once a valid baseline view has been detected and augment the training data with off-centre baseline views so that modest camera pans are still classified correctly.



Figure 6: Predicted: Invalid



Figure 7: Predicted: Valid

5.2. Classical Court Corner Detection Accuracy

Qualitatively the initial centroid method proved effective on clean hard-court surfaces. However, this method proved inconsistent across diverse datasets, particularly on clay or grass courts where line contrast and texture vary. A subsequent attempt used colour and brightness masking in a pre-processing stage to isolate high-intensity white regions representing court lines. Identifying centroid clusters near the image centre enhanced visibility but remained sensitive to lighting and surface wear as seen in figure 8.



Figure 8: Cropping Approach

With the collected results compared to the ground truth datasets, we arrive at the table of summary statistics below. As seen in the methodology section (Figure 8), we can anticipate that the canny-hough based cropping approach to be inconsistent across frames. This is mainly due to the difficulty of pre-processing and fine tuning a single threshold value across a diverse range of data.

As seen in the Australian Open trial dataset, we can easily see that the centroid based approach performed well, however it wasn't able to find corners for other trial frames for the US Open and Wimbledon - this is largely due to the thresholding constraints and most definitely the centroid based cropping approach. Never the less the LSD method proved successful at finding corners but it was not accurate as documented in Table 3.

More valuable results are qualitative to understand the behaviour across differing datasets. Figure 9 demonstrates that the baseline is accurately detected however the crowd is and other white parts are used as placeholders instead of the court corners. This is due to the line thickness, as the cameras perspective of the court exaggerates the far line contrast, small pixel heights may fall below LSD's gradient threshold and not be detected.



Figure 9: LSD US Open Results

NB: In this report we believe it is important to show the classical methods pitfalls and inaccuracies and our efforts.

These methods could most definitely been improved by selecting a more appropriate colour space such as HSL. This allows tennis court line detection to be more robust to lighting and surface variations by separating brightness (lightness) from colour information. By thresholding high-lightness and low-saturation regions, white court lines could’ve been isolated consistently across different court types and illumination conditions.

5.3. Hybrid Court Corner Detection

As shown in Table 2, the hybrid model demonstrates high spatial consistency and accuracy across 8,841 frames, with standard deviations for all corner positions ranging only between 27 and 33 pixels. The predicted corners are well aligned horizontally, with the baseline corners averaging around a y-coordinate of 560 px, and the returner’s line (top baseline) around 188 px, indicating a stable vertical structure across frames. Furthermore, the x-coordinates follow the expected camera perspective, where the top and bottom left corners differ by approximately 300 pixels, as do the top and bottom right corners. This consistent horizontal offset reflects the natural convergence of court lines toward the far baseline, confirming that the model accurately preserves the projective geometry of the broadcast viewpoint.

Table 2: Hybrid Model Summary Statistics

Point	Count	Mean _x	Std _x	Mean _y	Std _y
Baseline Top Left	8841	401.8	40.7	188.6	27.6
Baseline Top Right	8841	878.8	45.9	188.7	27.6
Baseline Bottom Left	8841	191.9	63.8	559.1	32.8
Baseline Bottom Right	8841	1088.8	67.7	559.4	33.2

5.4. Player Tracking

The MOG2 background subtraction was very effective and robust with the stable camera assumption as seen in Figure 10. The aspect ratio constraint ensured that tracked blobs resembled human-like dimensions however it was noted that in some rallies players exhibit such an outstretched reach for the ball that the condition is not met.

The Wimbledon womens (W) dataset achieved superior performance with an RMSE of 4.05m, while the Australian

Open male (M) dataset exhibited significantly higher error (RMSE: 13.5m) largely due to the more stark contrast and brightness evident on the Wimbledon court, creating a more robust foreground segmentation. We utilised YOLO as our ground truth under the assumption of SOTA performance. YOLO however, as a ground truth does have inconsistencies in predictions, analysis of the coordinate distributions reveals underlying deviations in predictions. We noticed that there was extreme y-axis variance in the M dataset ($\sigma_y = 1217.6\text{px}$) indicating substantial vertical tracking instability, likely from crowd detections and parallax depth plane distortion in the upper frame regions.

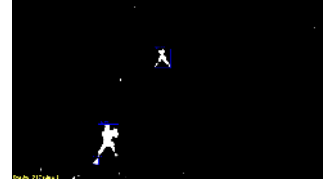


Figure 10: MOG2 Background Subtraction

Our MOG2 approach demonstrated more constrained spatial distributions ($W: \sigma_x = 207.3\text{px}, \sigma_y = 149.7\text{px}$; $M: \sigma_x = 314.2\text{px}, \sigma_y = 184.2\text{px}$), particularly in the vertical axis, reflecting the motion mask’s effectiveness at isolating tennis court activity compared to a pre-trained model. The motion-based segmentation actively filtered the false detections, as stationary crowd members and ball persons contributed minimally to the motion mask. However, this strength exposed a fundamental limitation in the initialisation delay, During rally starts, when players were stationary waiting for the serve, MOG2 produced zero detections until motion commenced, while YOLOv8 maintained tracking.

A notable challenge emerged in player identity persistence. Both methods tracked the two highest-confidence detections, causing frequent label alternation between players. As the two players were detected and labelled purely from the motion mask and geometric constraints, it proved difficult to integrate semantic understanding of the detections, such as *remembering* which player was which thus making logging and comparison unstable.

The method’s robustness degraded under camera motion. Since MOG2 maintains per-pixel colour intensity models, subtle camera rotation or zoom introduced undesirable artefacts such as court boundaries and crowd regions into the foreground mask (Figure 13), violating the static camera assumption. To improve our design, we can use the detected court corners such that only movement within and around this region is considered. This post-processing filter can remove crowd and background noise, and combined with our already existing area and aspect ratio thresholding we can ensure only realistic player-sized detections are retained.

5.5. Ball Tracking

Our tracker achieved a mean positional error of 76px which while acceptable compared to our trained YOLO model on the mens dataset, many false positives were observed, indicating spurious detections and consequential tracking failure. It must be noted that our self trained YOLO model was also not as robust as initially expected, attesting to the difficulty of ball tracking and getting a robust ground truth. On our hand labelled dataset, it achieved errors less then 50px proving that in ideal conditions, the tracker worked well.

The primary limitation stemmed from the ball's small size, low contrast and instability of the morphological operations, making it sometimes indistinguishable from noise artefacts and blob features on players, rackets and surrounding court objects leading to minimal tracking overlap with the ground truth. In an attempt to mitigate the aforementioned false positives, the player subtraction was an effective mitigator but costly solution that inherently loses the ball during overlapping player-ball proximity.

The kalman filter prediction provided temporal continuity but struggled with tennis-specific motion characteristics such as direction changes, fast serves, and variable directional movement during rallies. The tracker performed adequately in frames with clear ball visibility and predictable trajectories (ball movement between players), but degraded under motion blur, occlusion behind the net or player, and rapid, unpredictable velocity transitions. Unlike player tracking where motion segmentation inherently filtered background, ball tracking lacked a comparable discriminative feature, resulting in frequent spurious detections that no combination of morphological kernels or thresholds could reliably eliminate.

We selected data such to detect the ball in ideal, non occluded frames. However overall, the ball tracking suffered due to the low native frame rate of the footage, introducing motion blur such that the pipeline could not reliably track the ball across frames. We believe our model could reliably track the ball with a higher frame rate and resolution.

5.6. Pipeline Results

The complete tracking pipeline met its design objectives, demonstrating operation across all datasets. Processing only begins once a valid frame is confirmed by the classifier. Court detection is triggered once per valid rally, maintaining consistent perspective transformation and avoiding redundant computation. Qualitatively, during invalid frames, down-stream processes are paused and re-initialised only when valid input is detected. Visual inspection, seen in Figure 11 & 12, confirmed accurate player localisation, stable identities, and geometrically correct bird's-eye projections, with ball detection performing reliably when visibility and motion assumptions are fulfilled (See Link A).

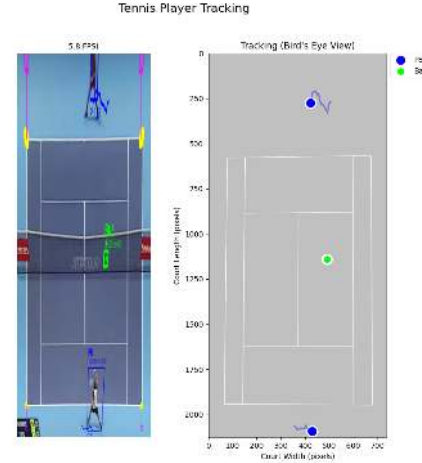


Figure 11: Visual Overlay of Complete Tracking Pipeline

6. Conclusion

The pipeline achieved moderately robust player and ball tracking from single-camera footage, meeting design objectives through integrated modules starting with frame validation. Classical methods provided stable court calibration and player tracking, while a lightweight pre-trained corner detector improved robustness. Ball tracking was effective under clear visibility but remained sensitive to occlusion and motion blur. Overall, the system shows that low-cost, single-camera vision pipelines can deliver meaningful tennis analytics, with future work aimed at improving real-time speed and tracking robustness.

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A. Full Pipeline

Full pipeline demonstration available at:

<https://youtu.be/GMS0oGpxVDY>



Figure 12: Full pipeline

B. Corner Detection Errors

Table 3: Corner Detection Error Summary Across Tournament Datasets

Dataset	Method	TL Err (px)	TR Err (px)	BL Err (px)	BR Err (px)	Mean $\pm$ SD (px)
2 nd Australian Open	Canny-Hough	1084.4	300.6	1506.2	28.2	729.8 $\pm$ 592.7
	LSD	329.8	1009.9	1503.6	418.0	815.4 $\pm$ 475.7
2 nd US Open	Canny-Hough	—	—	—	—	Failed
	LSD	393.9	1111.5	1577.8	6.1	772.3 $\pm$ 611.2
2 nd Wimbledon	Canny-Hough	—	—	—	—	Failed
	LSD	94.4	1279.1	1676.7	10.3	765.1 $\pm$ 727.1

C. Conversion Equations

Real-world coordinates for tracked players and ball are calculated via:

$$x_{metres} = (x'_{player} - p) \cdot s_x, \quad y_{metres} = (y'_{player} - p) \cdot s_y \quad (2)$$

D. Camera Movement

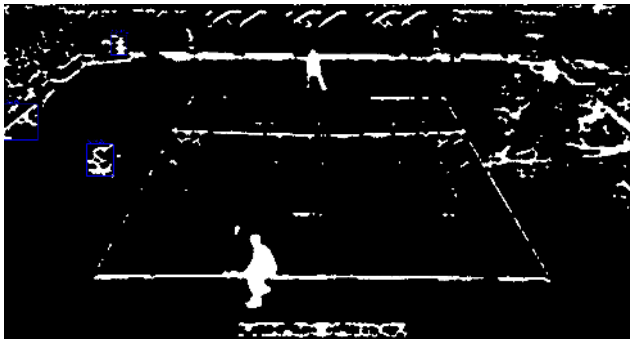


Figure 13: Camera Movement

E. Frame Classification Features

The following features were extracted to determine if a given frame was valid or invalid.



Figure 14: "Net proxy" (horizontal band energy)



Figure 15: Line segment detection



Figure 16: Orientation structure

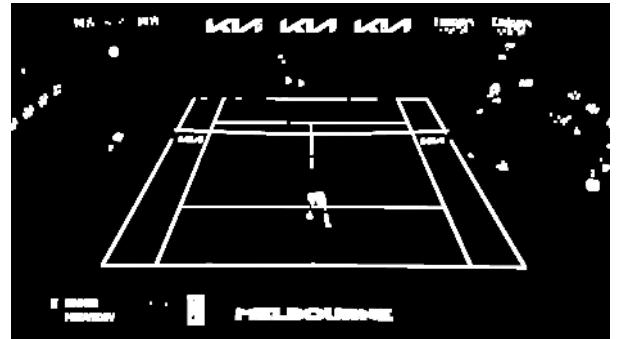


Figure 17: White line mask

Statement of Contributions

AMME5710 – Major Project

Autonomous Tennis Analysis

This document outlines the individual contributions made by each team member across the various components of the project, including research, design, implementation, evaluation, and report preparation.

Summary of Individual Contributions

Name	Description of Contributions
Sam Duckworth	• Initial court corner detection experiments and early testing. • Developed code modules for player/ball tracking, warping/visualisation, testing and analysis. • Integrated all modules into a complete working pipeline and did full system testing. • Wrote sections on player, ball tracking, visualisation/warping & general report writing. • Designed and prepared presentation slide deck.
Owen Dyne	• KNN-based classifier + feature extraction for gameplay vs auxiliary footage. • Dataset curation and labelling for the scene classifier. • Wrote relevant report sections for the classifier and data. • Edited report and wrote abstract
Ari Stathis	• Designed and implemented classical court-corner detection models and relevant report sections • Further trials on classical court detection models. • Initial report abstract and introduction • Full report write up for initial starting point for refinement • Methodology write-up and literature review

Signatures



Sam Duckworth



Owen Dyne



Ari Stathis

Date Submitted: 2025-11-26